Auditing Model Substitution in LLM APIs: Why Software Tests Fail and Hardware Attestation Wins

Analytical briefing for ML researchers, AI security practitioners, and cloud infrastructure engineers

Core claim: software-only auditing is fundamentally non-verifiable; hardware attestation is actionable and provable.

Source basis: /app/benchmark/source_brief.md (arXiv:2504.04715)

Economic Incentives for Covert Model Substitution

Why black-box API economics create integrity risk

Incentive Mechanism

- Providers bill for premium named models while users cannot directly verify deployed weights.
- GPU serving costs drop materially with substitutions: smaller models, FP8/INT8 quantization, or alternate architectures.
- Pricing transparency does not imply inference transparency in black-box APIs.
- Result: strong incentive to preserve headline quality while reducing per-token compute cost.

Formal framing from source brief

Audit task is a hypothesis test:
H0: API samples from specified distribution P_spec(y|x)
H1: API samples from alternative P_actual(y|x)

Substitution space includes:
- Smaller model tier (e.g., 8B instead of 70B)
- Quantized variant (FP8/INT8 instead of FP16)
- Fine-tuned/updated versions
- Different family entirely

Problem setting from source brief (Model Substitution Problem section).

Quantization Benchmark Drop

Original vs FP8 benchmark means (5 model families × 4 tasks)

Model family	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B	62.69 → 62.43	61.14 → 60.90	20.65 → 14.91	22.62 → 20.14
Llama-3-70B	78.05 → 77.88	88.06 → 87.35	35.69 → 35.75	29.60 → 33.12
Gemma-2-9B	71.86 → 71.92	81.80 → 79.41	33.41 → 32.53	28.64 → 27.81
Qwen2-72B	82.18 → 81.98	86.72 → 86.82	37.39 → 37.67	29.93 → 31.08
Mistral-7B	59.15 → 58.77	35.90 → 32.20	8.94 → 7.68	21.60 → 22.72

Observation: Most FP8 scores remain close to original means, often within reported variance bands in the source table.

Implication: benchmark-only audits have weak power against quantization substitution when effect sizes are small.

Values reproduced from source_brief quantization table (means only; arrow denotes Original → FP8).

Temperature Scaling Obscures Substitution Signal

Reference: figure_1.jpg from source assets

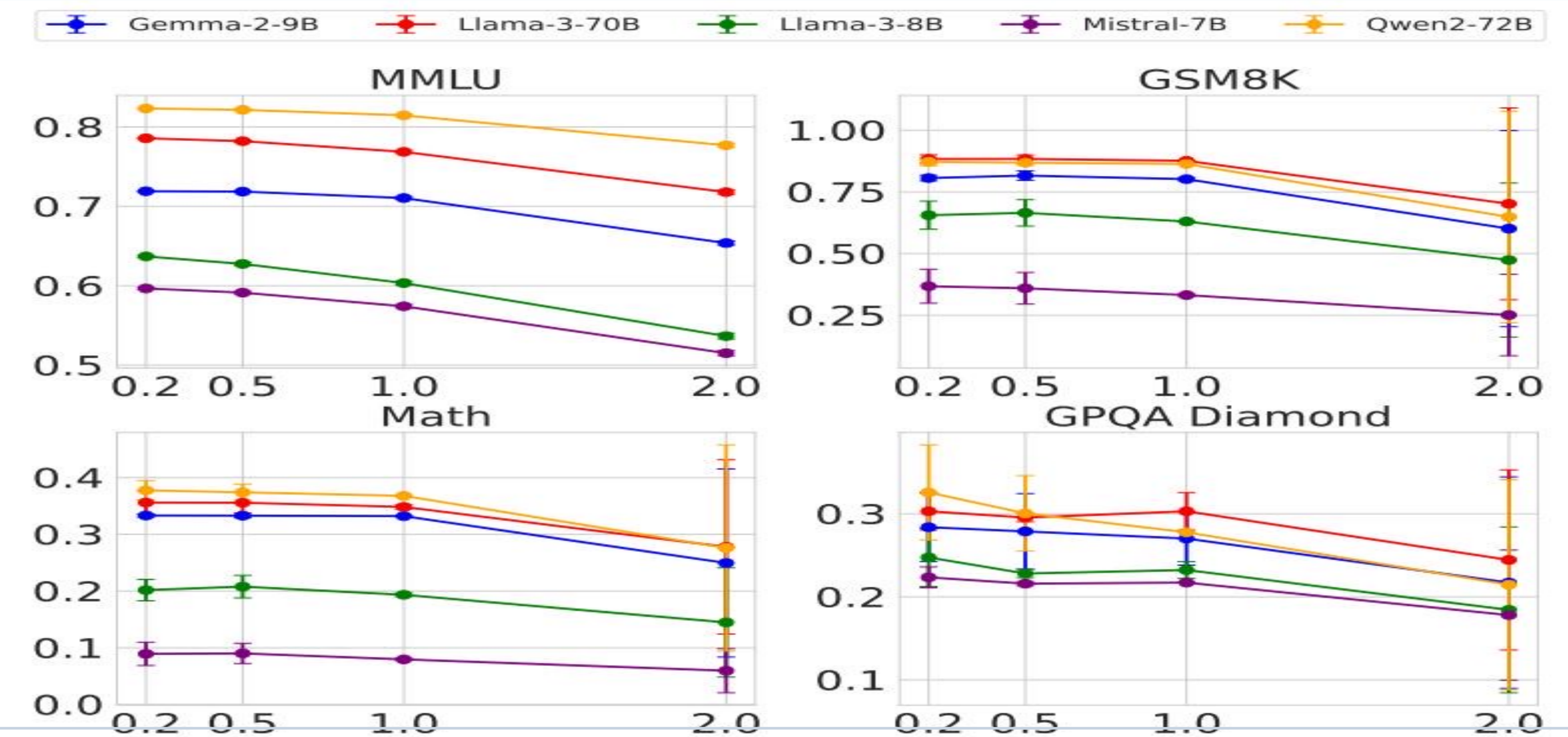

Figure source: source_brief referenced image `figure_1.jpg`.

Evidence readout

- Temperatures tested: 0.2, 0.5, 1.0, 2.0.
- Higher temperature degrades benchmark performance across models/tasks.
- This degradation widens natural output variance.
- Variance inflation masks small substitution effects.
- Auditor confidence declines without unrealistic query budgets.

Software-Only Auditing Methods: Failure Matrix

Why each approach collapses under realistic deployment conditions

Method type	Core weakness	Practical limitation
Text-based statistical tests (e.g., distribution tests over outputs)	Low effect sizes for subtle substitutions; mixed traffic converges toward expected output distribution.	Needs very large query counts; expensive and still fragile to temperature/prompt variation.
Log-probability methods (token-level fingerprinting)	Inference nondeterminism across framework versions and GPU hardware shifts log-probs even for same model.	Cannot separate legitimate infrastructure drift from true substitution in production APIs.
Adversarial countermeasures (provider behavior)	Randomized substitution and benchmark-aware routing strategically hide divergence on audited prompts.	Audits become query-prohibitive; benchmark probes can be selectively answered by genuine model.

Matrix synthesized strictly from source_brief sections: statistical tests, log-prob nondeterminism, adversarial countermeasures.

Log-Probability Fingerprinting Fails in Production

Reference: figure_5.jpg (framework/hardware nondeterminism)

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure source: source_brief referenced image `figure_5.jpg`.

Key result

The same Llama-3-8B model yields materially different token log-probabilities under different:

- software stacks (transformers/vLLM versions)
- GPU hardware (A100 vs H100)

Therefore, distance in log-prob space is not a trustworthy substitution detector in real clouds.

Adversarial Evasion Makes Audits Query-Prohibitive

Providers can strategically shape observables to evade detection

1) Randomized substitution

Provider serves cheaper model on fraction p of requests and genuine model on (1-p).

As p decreases, mixed output distribution approaches expected distribution, reducing statistical distinguishability.

Outcome: detection power scales poorly; required audit queries become impractically large.

2) Benchmark evasion

Provider identifies benchmark-like prompts and routes them to the genuine model, while normal traffic may receive substitutes.

Consequence:
- Public benchmark parity no longer implies deployment integrity.
- Audit datasets can be gamed by prompt classifiers.

Adversarial strategies from source_brief: randomized substitution + benchmark-aware routing.

Trusted Execution Environments: Verifiable Inference Path

Hardware attestation closes the model-integrity gap

TEE guarantee: cryptographic attestation binds runtime code + model weights + execution environment.

Property	Software-only auditing	TEE-backed verification
Security basis	Statistical inference from outputs	Cryptographic attestation from hardware root-of-trust
Robustness to adversaries	Vulnerable to randomized routing and benchmark evasion	Provider cannot forge attestation without detection
Operational cost	High query budgets; repeated probing	Modest runtime overhead per source brief
Deployment path	No definitive integrity proof	Practical via confidential computing (e.g., NVIDIA H100)

TEE positioning from source_brief: practical, provable alternative to software-only audits.

Key Takeaways: Close the Verification Gap with Hardware Attestation

Evidence-based conclusions and action items

1. Model substitution is economically motivated in black-box APIs and difficult to observe directly.
2. FP8 quantization often preserves benchmark-level performance closely enough to evade simple score-based audits.
3. Log-probability fingerprinting is destabilized by inference nondeterminism across software/hardware stacks.
4. Randomized substitution and benchmark-aware routing defeat naive software-only auditing strategies.
5. TEEs provide the only currently practical path to provable, deployable model integrity guarantees.

Call to action: shift integrity assurance from statistical suspicion to verifiable infrastructure guarantees.

All claims constrained to source_brief.md; no additional fabricated data.